

Use Case LM–1.2.1: Initialize Launch Mechanism

Scope: Launch Mission

Primary Actor: Flight Director

Secondar Actor: None

Related Use Cases: *Use Case 1 Initialize Launch,*

Use Case 1.1 Initialize Rocket

Use Case 1.2 Initialize Launch Platform

Stake Holders & Interests:

- Flight Director: Wants to set up a Launch via initialilzing all the Systems and Sub-Systems including:
 - Launch Platform
 - Launch Mechanism
- Mission Planner: Wants the Mission Initialized with the appropriate Mission Parameters loaded

Pre-Conditions: The System is running and ready to initialize and monitor the appropriate Systems and Sub-Systems

Post-Conditions: The Launch Mechanism is initialized with the appropriate parameters.

Flight Director	System
1. Enters the Mission Parameters File (See <i>Use Case 1 Initialize Launch</i>)	
	2. Opens the Mission Parameters File
	3. Finds the Launch Platform Data
	4. Finds the Launch Mechanism Data in the Launch Platform Data
	5. Stores the Launch Mechanism Data
	6. Finds the Rocket Data
	7. Stores the needed Rocket Data
	8. Returns the Initialized Launch Mechanism Data
9. Receives the Initialized Launch Mechanism Data	

Extensios (or Alternative Flows)

- 2a. If the System cannot open the Mission Parameters File, then the System alerts the Flight Director of the error. Flight Director determines next course of action.
- 4a. If the System cannot access the Launch Mechanism Data, then the System alerts the Flight Director of the error. Flight Director determines next course of action.

- 6a.,7a. If there is no Rocket Data available, then the System continues without storage of the Rocket Data. System alerts the Flight Director.

Notes/Constraints

- *a. This Use Case documents the Launch Mechanism Initialization, not the Mechanism Monitoring (See *Use Case 4.1 Monitor Initialization*)—this includes monitoring System Measurements of the force and downward force.
 - *.b Upon receipt of an alert, the next course of action for the Flight Director is aborting the mission or choosing another Mission Parameter File.
- 6b.,7b. Since prior Rocket Initialization is a definite possibility, a more likely scenario is the System accesses the initialized Rocket Data.

Special Requirements:

- None

Technology & Variations List:

- There could be many types of Mission Parameters Files in different formats. The System needs to account for all the different types of files and read the accordingly.
- Regardless of the type of file, the System Data will follow a Standard.
- The Launch Platform may have no Launch Mechanisms.

Frequency of Occurrence

- Every time a mission is planned or simulated.

Open Issues

- 6c.,7c. There are different possibilities if there is no Initialized Rocket Data (See *Use Case 1.1 Initialize Rocket*) including
- Giving the Flight Director the option to initialize.
 - Initializing the System without a Rocket connected to the Mechanism